Changing basis for matrices of linear maps

Soham Saha
https://soham-saha.github.io

Take basis $\{u_1, u_2, u_3\}$ for $\mathbb{R}^3$, and basis $\{v_1, v_2\}$ for $\mathbb{R}^2$. We shall have the "formal" (because entries are vectors) row matrices

$$U := \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}$$
$$V := \begin{bmatrix} v_1 & v_2 \end{bmatrix},$$

which store the ordered basis data for $\mathbb{R}^3$ and $\mathbb{R}^2$.

Given a map $\mathcal{T} \in \mathrm{Hom}_{\mathbb{R}}(\mathbb{R}^3, \mathbb{R}^2)$, suppose the matrix of $\mathcal{T}$ with respect to the basis U and V looks like

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Observe that

$$VA = \mathcal{T}(U),$$
$$\text{where } \mathcal{T}(E) := \begin{bmatrix} \mathcal{T}(u_1) & \mathcal{T}(u_2) & \mathcal{T}(u_3) \end{bmatrix}.$$

Case 1 : Column operations on A

Suppose we do the column operation $C_2 \to C_2 + 2C_1$ on A, which corresponds to multiplying A on the right by $M_1 = \begin{bmatrix} 1 & 2 & \\ & 1 & \\ & & 1 \end{bmatrix}$, so we get

$$AM_1 = \begin{bmatrix} 1 & 4 & 3 \\ 4 & 13 & 6 \end{bmatrix} = \text{result of doing } C_2 \to C_2 + 2C_1 \text{ on } A,$$

and $VA = \mathcal{T}(U)$ becomes

$$VAM_1 = \mathcal{T}(U)M_1 = \mathcal{T}(UM_1), \text{using linearity of } \mathcal{T},$$

which tells us that AM_1 is the matrix of $\mathcal{T}$ with respect to the "new basis" $UM_1 = \begin{bmatrix} u_1 & u_2 + 2u_1 & u_3 \end{bmatrix}$ in the domain and same old V in the codomain.

Case 2 : Row operations on A

Suppose we do the row operation $R_2 \to R_2 + 2R_1$, which corresponds to multiplying A on the left by $M_2 = \begin{bmatrix} 1 & \\ 2 & 1 \end{bmatrix}$, so we get

$$M_2A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & 9 & 12 \end{bmatrix} = \text{result of doing } R_2 \to R_2 + 2R_1 \text{ on } A,$$

and $VA = \mathcal{T}(U)$ becomes

$$(VM_2^{-1})M_2A = \mathcal{T}(U),$$

which tells us that M_2A is the matrix of $\mathcal{T}$ with respect to the basis U in the domain and basis $VM_2^{-1} = \begin{bmatrix} v_1 - 2v_2 & v_2 \end{bmatrix}$ in the codomain.

This process helps us keep track of how the basis gets updated as we perform row and column operation on the matrices.

Observe that if $D = Q^{-1}AP$, then

$$VA = \mathcal{T}(U) \text{ becomes } (VQ)Q^{-1}AP = \mathcal{T}(UP) \Rightarrow (VQ)D = \mathcal{T}(UP)$$

This tells us that D is the matrix of $\mathcal{T}$ under the new basis UP in the domain and VQ is the codomain.